REACT Micro-Randomized Trial
Analysis Specification & Decision

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1. PRIMARY BEHAVIORAL OUTCOMES (§15.0) & SECONDARY PROXIMAL OUTCOMES

Construct	Candidate Item(s)	Scale	Recall Horizon	Operationalization & Estimator
Primary Behavioral Domain Outcomes — Time-locked outcome window [+60 min, +2 h] post-decision				
Posting urge	`B7_urge`	Likert (1–7)	Now / Current	Linear WCLS; primary excursion effect [1]
Online behavior	`B7_posted`	Binary	Past few hours	1 = Yes, 0 = No; Qian et al. Binary MRT [2]
	`B7_charged`	Likert (1–7)	Past few hours	Linear WCLS excursion model
	`B7_self_stopped`	Categorical	Past few hours	3-level categorical choice item
Alcohol behavior	`B6_consumed`	Binary	Today	1 = Yes, 0 = No; Qian et al. Binary MRT [2]
	`B6_drink_count`	Count (≥ 0)	Today	Discrete count/rate GLM excursion model
Alcohol urge	`B6_urge`	Likert (1–7)	Now / Current	Linear WCLS [1]
Eating (emotional)	`B4_emotion_influence`	Likert (1–7)	Since last check-in	Linear WCLS among `B4_ate` = Yes
Uncontrolled eat.	`B4_loss_of_control`	Likert (1–7)	Since last check-in	Linear WCLS among `B4_ate` = Yes
Secondary Proximal Affective Outcomes				
Affective valence	`B1_valence`	Scale (1–7)	Now / Current	1 = Very Unpleasant, 7 = Very Pleasant; WCLS
Perceived stress	`B2_stress`	Scale (1–7)	Now / Current	1 = Not at all, 7 = Extremely stressed; WCLS

2. CONFIRMATORY TIER-1 MODERATORS & TIME

- **Baseline Trait Inventories:**
  - **SUPPS-P:** Negative & Positive Urgency subscales.
  - **DMQ-R:** Coping Drinking Motives subscale.
  - **TFEQ-R18:** Emotional & Uncontrolled Eating.
  - **DERS-16:** Difficulties in Emotion Regulation Total.
  - **MAIA-2:** Noticing & Body Listening subscales.
- **Temporal Moderators (Within-Person):**
  - **Hour of Day (S_{it}^{hour}):** Timestamp (ET), centered at 15:00.
  - **Study Day (S_{it}^{day}):** Day in study, centered at mid-study; evaluates attenuation/habituation across Weeks 2–5.
- **Moderator Adjustment:** Main effects of S_{it} are strictly included in control terms $g(H_{it})$ to preserve orthogonal excursion treatment contrasts.

3. ESTIMATOR SELECTION: BINARY VS. LIKERT

- **Binary outcomes** (`B7_posted`, `B6_consumed`): Weighted and centered risk difference as primary; log-RR for relative risks [2].
- **Likert items** (all 1–7 scales): Linear WCLS applied uniformly as primary robust model [1]; ordinal estimators reserved for sensitivity checks.

4. COPING VS. ACTIVE CONTROL SUB-STUDY

- **Two-Stage Randomization:** At eligible decision points ($I_{it} = 1$), prompt delivery is drawn ($A_{it} \in \{0, 1\}$, $p = 0.5$). If $A_{it} = 1$, arm allocation is randomized ($p_{\text{arm}} = 0.5$): Coping JITAI ($M_{it} = 1$) vs. Active Control ($M_{it} = 0$).
- **Causal Estimand:** Isolates active skill content from alert reactivity by evaluating the conditional mean difference between message arms among prompted decision points.
- **Estimation:** WCLS on the $A_{it} = 1$ sub-sample, centered on $(M_{it} - \tilde{p}_{\text{arm}})$.

5. DETECTABLE EFFECT SIZES

- **Parameters:** $N = 80$, 28 study days, 6 DPs/day, true MSSD availability ≈ 0.21 (production system check), $\alpha = 0.05$ (two-sided), 80% power.
- **Sub-Study Scaling:** Evaluating $A_{it} = 1$ halves effective sample size, scaling the detectable effect size by $1/\sqrt{p_{\text{send}}} \approx \sqrt{2} \approx 1.414$ relative to the main trial excursion effect.

References

[1] A. Boruvka, D. Almirall, K. Witkiewitz, S. A. Murphy. *JASA*, 113(523):1112–1121, 2018.

[2] T. Qian, H. Yoo, P. Klasnja, D. Almirall, S. A. Murphy. *Biometrika*, 108(3):507–527, 2021.